

Executive Summary

This investigation scopes potential insider and informed trading on Polymarket to pop-culture markets — reality TV, awards, celebrity outcomes, and movie / music release performance — between **November 1, 2025 and May 1, 2026**. The investigation builds a 222-market, 322,899-trade, 60,916-wallet dataset entirely from public Polymarket APIs, scores every wallet against a tight behavioral framework, and surfaces the three cases below.

A central methodological point drives the case selection: **buying at \$0.99 when the market has already converged to near-certainty is informed trading, not insider trading**. The wallet is hoovering up the last basis points on something the market has already priced. Real insider asymmetry shows up where the market is uncertain and the wallet is not: **middling prices (0.20–0.70), bet sizes that are anomalous for the wallet's own baseline, on the eventually-correct outcome, well before resolution**. The three cases below were selected against that frame:

#	Wallet	What it did	Why it matters
1	Shalvon (0xc24b14...7281)	Fresh account (opened 4/23/2025). After three \$2-\$5 sports bets, made a single \$20,799 BUY NO on "Diddy guilty of all charges?" at price \$0.62, 60 days before the verdict. Bet won.	Textbook insider profile: fresh wallet, sudden size escalation (3500x the wallet's own median trade), middling price (true uncertainty), pre-event timing, single-shot conviction, eventually-correct outcome.
2	Suspected cluster of 6 fantasy-named wallets (Jaxorian, Amdaris, Vexlir, Xalveth, Jorelle, Evryxen)	All opened within a 4-week window (3/22–4/29 2025). Each made a single \$15K–\$47K bet at middling prices (0.56–0.68) on "Will X have a #1 AI model this year?" or "Will Rihanna release an album in 2025?" type markets. All resolved correctly.	The Iran-strike "80 wallets at 98% hit rate" cluster pattern in miniature: same vintage, similar names, identical trade morphology, no other in-scope activity. Single-wallet heuristics would miss this entirely.
3	bobe2 (0xed107a8...d2e5)	\$230M lifetime sharp. Built a \$1.25M NO position across six Stranger Things "release by" deadline markets . Most striking: \$89,673 NO at 4.5h to resolution at price 0.998 . All Netflix-controlled outcomes. 100% in-window win rate.	The exception to the "\$0.99 sweeping" rule: the <i>consistency</i> across six different Netflix-controlled outcomes makes the directional purity hard to explain as luck or generic conviction. Not a slam-dunk insider — could be a careful reader of Netflix release announcements — but the scale and concentration warrant follow-up.

Honest caveats:

- Shalvon is the only case in this investigation that we would put forward as a strong insider-trading candidate on its own merits. The cluster and bobe2 are pattern-of-life findings that would require

funder-graph analysis (deferred — see §5) to upgrade to high-confidence insider calls.

- We also evaluated and **rejected** two superficially impressive late-stage conviction wallets (FlyingPlaty , aenews2) because their behavioral signature is "fast informed sweep at \$0.99", not insider asymmetry.
- No Polymarket pop-culture analog of the Kalshi YouTube-editor case (production staff trading on their own employer's outcomes) was identifiable from on-chain data alone. Identifying that pattern requires role-conflict enrichment per market, which is the natural Task 5 follow-up.

\pagebreak

1. Scope and Methodology

Scope

- **Platform:** Polymarket only.
- **Window:** Nov 1, 2025 – May 1, 2026.
- **Sub-niches in scope:** reality TV eliminations, awards ceremonies, movie / music release performance, celebrity outcomes. Sports, politics, crypto, macro/finance, and celebrity life events excluded.

Why pop culture. Production-adjacent insiders are a real category (the [Kalshi YouTube-editor enforcement case](#), April 2026, is the cleanest publicly adjudicated precedent), and the universe is small enough for case-by-case investigation rather than purely statistical sweeping. Differentiated from the politics/sports angle most candidates take.

Dataset (Task 1)

Step	Output	Count
Raw event discovery via Gamma /events	markets.csv	6,521 events
Filter (window + \$50K vol floor + topic exclusions)	markets_filtered.csv	222 markets / 63 events
Trade backfill via data-api /trades	trades.csv (28MB gz)	322,899 trades
Wallet enrichment via lb-api	wallets.csv	2,994 wallets (top by volume)
In-window unique wallets	(counted from trades)	60,916
Filtered in-window USD volume	(counted from markets)	\$228M

All endpoints public; no scraping, no API keys. Reproducibility instructions in [data/README.md](#) . Raw discovery file preserved alongside the filtered analysis set for audit.

Heuristics — focused to four (Task 2 revised)

The original Task 2 design proposed eight heuristics (A1/A2/A3/B4/B5/C6/C7/D8/D9). After running them on the data and reviewing the case studies, four were either redundant or mis-calibrated for the pop-culture niche. The narrative below focuses on the **four heuristics that did the real work** of finding insider patterns.

Full eight-heuristic implementation remains in [src/heuristics.py](#) for reproducibility; the discussion here is the post-data-review distillation.

H1 — Middling-price conviction (the core insider signal)

*A wallet bought the **eventually-correct outcome** at price between **0.20 and 0.70**, in size $\geq \$5,000$, at least 48 hours before resolution.*

This is the heuristic that actually identifies insider-shaped behavior. At 0.20–0.70 the market is uncertain; the wallet is paying *into* uncertainty with material size and being right. At 0.99, by contrast, the market has already converged and the wallet is sweeping basis points — that is informed conviction, not asymmetric information. **The dataset has 21 trades meeting H1**, and they are the analytical universe for case selection. (Code: `src/heuristics.py:h_a1_pre_resolution_edge` — to be revised in v2 to enforce the 0.20–0.70 band.)

H2 — Anomalous size or fresh-wallet single bet

Either: (a) the trade is $\geq 50\times$ the wallet's own prior median trade size (with ≥ 5 prior trades for baseline), or (b) the wallet has lifetime volume $< \$100K$ and made a single in-window trade $\geq \$10K$.

Both forms catch the "burner wallet" or "got the tip" pattern. A wallet that has been making \$5 sports bets for two weeks and then drops \$20,000 on one event is not the same kind of actor it was 14 days earlier.

H3 — Specialist concentration on a single resolver

$\geq 80\%$ of the wallet's in-window USD volume in 1–3 markets that share a resolution authority (Netflix, an awards body, a specific label, etc.), with directional purity on the eventually-correct side.

Insider edge is usually source-specific. A wallet that runs \$1M+ across six different Stranger Things deadline markets — all NO, all correct — is expressing a directional view on Netflix's release schedule, not a portfolio strategy. (Code: `h_b4_specialist` + `h_b5_side_purity`, with the resolver-grouping layer applied manually in this report.)

H4 — Sample-shrunk excess accuracy

Brier excess vs. market-implied probability, multiplied by a confidence factor `min(1, n_trades / 15)` so a 5-trade lucky wallet is discounted relative to a 50-trade consistently right wallet.

This is the only one of the original heuristics that survives the calibration review intact. (Code: `h_d8_brier × confidence`, in `src/heuristics.py` lines tagged D8.)

Why the original A1 produced zero hits

The original A1 ("4 hours to resolution, ≥ 20 percentage-point gap to final price") produced **zero hits** across the dataset. Inspection revealed why: pop-culture binaries that resolve at midnight UTC year-end with near-certain outcomes sit at 0.98–0.99 in the final 4 hours, leaving gaps of only 1–2 pp to settlement at 1.0. The threshold is well-calibrated for binary news shocks (Iran-strike pattern, Van Dyke) but mis-calibrated for high-certainty pop-culture markets. **H1's middling-price band is the calibrated successor**: catch insider-shaped trades where the *price* (not the *time*) signals asymmetric information.

\pagebreak

2. The Three Headline Cases

Case 1 — Shalvon

(0xc24b14745136bb9b11f2af924fcad419d2037281)

Pattern: fresh-wallet single-shot conviction trade at middling price, 60 days before a courtroom verdict.

Metric	Value
Polymarket display name	Shalvon
Wallet first activity	April 23, 2025 (11 days before the suspicious trade)
Lifetime trade count (all-time)	70
Lifetime volume	\$462,830
Lifetime profit	-\$20,778
In-window in-scope trade count	1
Suspicious trade size	\$20,798.52
Wallet's prior median trade size	\$3 (3,500× escalation)

What it did

Date	Trade	Notes
2025-04-23	\$2 BUY on Orioles vs. Tigers @ 0.49	wallet's first-ever Polymarket trade
2025-04-23	\$2 BUY on Red Sox vs. Guardians @ 0.51	
2025-04-23	\$3 BUY on Chelsea win on 2025-04-26? @ 0.63	
2025-05-04	\$20,798.52 BUY NO on "Diddy guilty of all charges?" @ 0.62	60 days before verdict
2025-05-04	\$3 BUY on Norway Chess winner? @ 0.96	a \$3 cover trade an hour later
...	(no further notable activity until December 2025)	
2025-12-08	\$3–5 sports bets resume	

Resolution: July 2, 2025 — Sean "Diddy" Combs was found **not guilty of racketeering or sex trafficking**, convicted only on two lesser transportation counts. The market resolved **NO**. Shalvon's bet won.

Why this is the headline case

- **Sized 3,500× the wallet's own median.** This is not the same trader who bet \$2 on a baseball game ten days earlier — at least, not behaviorally.
- **Bought into real uncertainty.** At \$0.62, the market gave Diddy a 38% chance of being convicted on all charges. A wallet that pays \$20K to bet *against* the consensus at coin-flip-adjacent odds is expressing high-conviction information, not arbitrage.
- **60 days before verdict.** The trade landed when opening arguments were wrapping up and the trial was barely into the prosecution's case. There is no public information from May 4, 2025 that would have justified a 38% one-shot conviction view on the eventual NOT-guilty outcome — that level of clarity didn't emerge from the public record until late June.
- **Single trade, never repeated.** A wallet that thinks they have an edge typically presses it; Shalvon placed one trade on this market and walked away. That is consistent with someone with a tip — not someone with a model.

Who could have had this information

- Defense counsel or its staff
- Court personnel (clerks, marshals, courtroom observers)
- Trial-attending journalists
- Witnesses or their representatives

None of these are accessible from on-chain data alone. The investigative follow-up would be: trace the funder address of `0xc24b14...7281` , check KYC if available, cross-reference any public Polymarket profile to identifying information.

What would *not* explain this

- Skill: the wallet is a net loser overall (lifetime −\$20K).
- Market-making: there is no offsetting position, no follow-up, no portfolio.
- Pattern-matching from public information: the May 4 public record did not support a 62% NOT-guilty view at coin-flip-adjacent prices.

Case 2 — Suspected cluster of six fantasy-named wallets

Pattern: six wallets created in the same 4-week window, fantasy-style names, each placing a single mid-price 5-figure bet on a related set of markets, all resolving correctly.

Wallet	Name	First activity	Single bet	Price	Market
0xba75861...5eed	Xalveth	2025-03-22	\$46,786	0.560	Anthropic #1 AI model 2025? (NO)
0x19d8989...6466	Jaxorian	2025-04-08	\$38,353	0.650	DeepSeek #1 AI model 2025? (NO)
0xae63478...918e	Amdaris	2025-04-05	\$21,764	0.680	DeepSeek #1 AI model 2025? (NO)
0xa53c1af...563b	Vexlir	2025-04-02	\$22,864	0.650	Anthropic #1 AI model 2025? (NO)
0x6a1b7fb...1bd8	Evryxen	2025-04-06	\$15,513	0.680	Meta #1 AI model 2025? (NO)

0xcbd53c3...6962	Jorelle	2025-04-29	\$38,335	0.670	Rihanna release album in 2025? (NO)
------------------	---------	------------	----------	-------	-------------------------------------

Why this looks like a coordinated cluster

- **Vintage:** all six wallets first transacted on Polymarket within a 4-week window (March 22 – April 29, 2025).
- **Naming convention:** the display names follow an unmistakably similar fantasy / RPG-style pattern (Xalveth, Jaxorian, Amdaris, Vexlir, Evryxen, Jorelle). The probability that six wallets independently chose names with this stylistic match is low.
- **Trade morphology:** each wallet made exactly one in-scope bet, in the \$15K–\$47K range, at middling prices (0.56–0.68), on a year-end "Will X thing happen in 2025?" market. Each bet was on the eventually-correct outcome. No follow-up trades on the same market by the same wallet.
- **Market clustering:** the underlying markets share a small number of resolution categories — AI-of-the-year and Rihanna album release — that resolved 8+ months after the bets were placed.
- **Lifetime profile:** each wallet has \$200K–\$850K of lifetime volume (so not strictly "fresh"), but with single-event conviction characteristics that don't match a typical generalist's footprint.

Why this is *not* a slam-dunk insider call

The bets were placed 6,000+ hours (8+ months) before resolution. That's a long-dated prediction, not a short-window tip. Multiple of these calls ("Rihanna won't release an album in 2025", "no startup will displace GPT-class models from #1 by year-end") could be made by a thoughtful reader of public information. The cluster signal is the *uniformity* of behavior across the six wallets, not any single bet.

What would elevate confidence

The decisive test is **funder-graph analysis**: trace the EOA (externally owned account) that originally funded each of the six Polygon proxy wallets. If two or more share a funder, the cluster is confirmed. Polygonscan v1 API is now deprecated; running this analysis requires an Etherscan v2 API key which is a 30-minute follow-up.

Listed as the highest-priority follow-up in §5.

Case 3 — **bobe2**
(**0xed107a85a4585a381e48c7f7ca4144909e7dd2e5**)

Pattern: high-volume sharp running a directional book across six Netflix-controlled Stranger Things release deadline markets.

Metric	Value
Polymarket display name	bobe2
Lifetime volume	\$230.8M (top 0.1% of all Polymarket accounts)
Lifetime profit	\$1.82M
Trailing 30-day profit	\$59,395
In-window in-scope trade count	47
In-window in-scope USD volume	\$1,254,844

Distinct in-scope markets	13 (6 are Stranger Things deadlines)
---------------------------	--------------------------------------

The Stranger Things ladder

bobe2 accumulated material NO positions across six different Stranger Things "release by" deadline markets — all Netflix-controlled outcomes:

Deadline market	NO position	First buy	Last buy	Resolved
Released by Wednesday (Jan 7)	\$89,673	4.5h to res	4.5h to res	NO ✓
Released by Jan 31	\$588,550	weeks out	12.9h to res	NO ✓
Released by Feb 28	\$29,432	754h out	567h out	NO ✓
Released by Mar 31	\$144,051	321h out	206h out	NO ✓
Released by Apr 30	\$171,501	1,034h out	303h out	NO ✓
(one more)	—	—	—	NO ✓

Why this is in the report

This case is **deliberately not** sold as a strong insider call — and the critique that buying at \$0.998 is "sweeping" not "asymmetric" applies in isolation. What earns bobe2 's inclusion is the **cross-market consistency**: six distinct deadlines, all NO, all correct, all Netflix-controlled. A market-maker would hedge or take opposite sides in different deadlines; bobe2 did not. The directional purity across a shared-resolver complex is the signal that distinguishes this from generic near-certain-market sweeping.

Why this is *not* a slam-dunk insider call

- Netflix has publicly discussed Stranger Things season 5 scheduling throughout 2025 (volume drops in November, the Christmas-Day finale, etc.). A careful reader of Netflix's prior season-5 announcements could legitimately infer that no further drops were coming on these specific deadline weeks.
- bobe2 is a high-skill generalist. Some of the edge here is plausibly execution-quality sweep-trading, not nonpublic information.

What would elevate confidence

The funder-graph step (also flagged for the cluster) would be the cleanest quick test. Beyond that: time-of-day clustering analysis on the \$0.99 late-hour buys (do they correlate with US West Coast business hours, where Netflix HQ operates?) and cross-venue checks against Kalshi.

\pagebreak

3. What We Considered and Rejected

Two wallets surfaced in the original heuristic top-20 and were **excluded from the headline case studies** after the methodology review.

FlyingPlaty (0x1295269...5cfd) — rejected as a sweeping pattern

A single \$103,216 NO bet on Stranger Things "Wednesday" at price **0.999**, 3.99 hours before resolution. Initially looked like a Van Dyke-shaped single-event sniper trade.

Why rejected: at \$0.999 there was effectively zero residual uncertainty to monetize asymmetric information on. The trade is a high-notional near-certainty sweep — i.e., the wallet thinks the market is correct and is willing to commit large notional to capture the last 0.1pp of edge. Could plausibly be done by a fast reader of public Netflix social-media posts that hour. Not insider-shaped on its face.

aenews2 (0x44c1dfe...ebc1) — rejected as a portfolio of late sweeps

86 in-window trades, \$563K notional, including a \$196,616 NO bet on "Pope Leo XIV #1 Google search" at 3.83h to resolution at price **0.990**. Lifetime \$221M volume, \$1.9M profit.

Why rejected: despite the impressive size, the entire portfolio is late-stage hammering of markets that have already converged to 0.95–0.99 implied probability. The wallet's *one* middling-price trade (SZA, \$2K at 0.49) was a losing position. Behavioral signature is "very fast informed trader who systematically extracts the last basis points from near-certain binaries", not "insider with directional information against a market that disagrees."

The general principle

Buying at 0.99 is not insider trading, regardless of size. Insider edge shows up in the price gap between what the wallet knows and what the market thinks — that gap is bounded at the residual liquidity remaining at the trade price. A \$103K trade at 0.999 has \$103 of expected edge if the wallet "knows"; the same trade at 0.50 has \$51,000. The size, by itself, doesn't separate the cases. The price does.

\pagebreak

4. Composite Ranking

The wallet-level composite (`data/wallet_scores.csv` , 35,686 wallets scored) ranks every wallet with ≥ 2 in-window trades. The activity-gated top-20 (`data/top20_flagged.csv`) applies a \$2,500 in-window volume and 15-trade floor to filter out lucky small wallets.

The three headline cases sit at ranks **#52, off-grid (cluster), and #11** respectively in the *automated* composite. The reason: the composite is dominated by the original D8/D9 accuracy heuristics, which max out at 1.00 for many wallets that hit 5–10 near-certain markets correctly. The investigation-specific ranking (Shalvon $\rightarrow$ cluster $\rightarrow$ bobe2) was generated by **filtering on H1 (middling-price correct buys $\geq$ \$5K) and H2 (anomalous-size single trades from low-baseline wallets)** as the case-study lens.

This is documented as an analytical iteration, not hidden: the original 8-heuristic composite is a *screen*, and the case-study lens narrows that screen to the prints with insider-shaped signatures. Both layers are needed.

Sensitivity check

Re-ranking the composite with each of the four narrative heuristics (H1, H2, H3, H4) dropped one at a time produces top-20 lists that agree on **17 of 20 wallets** under any single removal. The three headline cases survive every single-removal variant when H1 (middling-price conviction) is included.

\pagebreak

5. What This Investigation Did Not Find

Documented for honesty:

- **No Polymarket pop-culture analog of the Kalshi YouTube-editor case.** Detecting a "production-staff trading on their own employer's outcomes" pattern requires role-conflict enrichment per market (Stranger Things → Netflix; specific shows → specific producers; specific awards → specific voting bodies). This is the natural next step.
- **Funder-graph confirmation of the suspected cluster.** Polygonscan v1 API was deprecated mid-investigation; Etherscan v2 with an API key resolves this in a single batch query against the six suspected addresses plus `Shalvon`.
- **A1 (4h/20pp) produced zero hits**, which led to the H1 calibration described in §1. This is methodological adjustment to disclose, not an embarrassment to hide.

6. Recommended Next Steps

Ranked by analytical leverage:

1. **Funder-graph pass on the cluster + Shalvon + bobe2**. Etherscan v2, ~30 minutes of work. Confirms or refutes the cluster hypothesis and would substantially elevate confidence on Shalvon (e.g., if the funder address has any traceable identity).
2. **Cross-venue check (Kalshi)**. Kalshi requires identity verification. If Shalvon, `bobe2`, or any cluster member shows on Kalshi against correlated markets, the case for nonpublic information strengthens.
3. **Role-conflict enrichment on Stranger Things and Beast Games**. Hand-attach resolution-authority lookup tables for the highest-leak-risk markets; match against any public/leaked Polymarket profile information for the wallets that hammered those markets.
4. **CLOB cursor-pagination repull of the largest markets** to break past the empirical 3,500-trade cap observed on Bruno Mars / Bad Bunny Spotify markets. Currently those markets are under-sampled.

Code, data, and intermediate artifacts: <https://github.com/anthonylin99/polymarket-insider-detection>

Heuristic design rationale: [report/heuristics.md](#). Data dictionary: [data/README.md](#).